MATH 20250 May 6, 2023
Prof. Lanier **Problem Swap 2** due: Wednesday, May 10, 2023

Solve at least two of the following problems. Post your solutions to the Discussions thread on Canvas by 11:59pm on the due date. It is fine if multiple people post solutions to the same problem. Also, this is not a race! The discussion thread supports the use of LaTeX, and using this is the preferred method for sharing your work. You can also upload a pdf or other file if needed. You may collaborate with others—this is encouraged!—but please note your collaboration on any solutions you post. Similarly, note if you use any outside resources. If you have any clarifying questions about a problem, please ask the the author about it in a polite and generous fashion.

Agustin Valenzuela Pidal:

Assume A is an $n \times n$ non-zero invertible matrix. Over $\mathbb{C}$, is it possible to have $\det(A) = \text{trace}(A)$? If the answer is yes, write a formula for A. If no, explain why.

Aidan Schmitt:

Let A and B be two 2×2 matrices with identical eigenvectors and eigenvalues c and d. What are the eigenvalues of AB and AB^{-1}?

Alan Rodal:

For what value of x does the matrix A $= \begin{pmatrix} 2 & x \\ 1 & 5 \end{pmatrix}$ have only 7/2 as an eigenvalue?

Andrew Mark Welch:

For what real values of k is the matrix $A = \begin{pmatrix} 0 & -2 \\ 2 & k \end{pmatrix}$ diagonalizable over $\mathbb{R}$? For what real values of k is it diagonalizable over $\mathbb{C}$?

Arina Oberoi: If matrix A has eigenvalue = 8, calculate the following eigenvalues:

- A^{-1}
- A^6
- $9A + 2$
- $A(I)$
- $1/A$
- $A + I + A^2$

Arjun Sohur:

Give an example of a subspace of 2×2 matrices with real elements where all elements are diagonalizable, but not invertible, if such a subspace exists. If a subspace with these properties does not exist, explain why.

Avery Fox:

Give an example of a square matrix (or matrices) that satisfy each of the following. If not possible, then why?

1. A projection matrix such that none of its eigenvalues are zero.
2. A matrix with determinant of 26, a trace of 20, and at least one eigenvalue with algebraic multiplicity greater than 1.
3. An upper triangular matrix and an anti-symmetric matrix that have the same eigenvalues but aren't similar.
4. An invertible matrix not equal to its inverse that has the same eigenvalues as its inverse.

Bonnie Ko:

What is the geometric interpretation of the determinant of a 2×2 matrix?

Brian Li:

Find an orthogonal basis for $\mathbb{R}^3$ that's not the standard basis (or a scaled version of it).

Hint: Two vectors are orthogonal if their dot product is 0.

Cellestine Harig:

Find the determinant and the trace of the given matrix.

$$\begin{pmatrix} 9 & 13 & 5 & 2 \\ 1 & 11 & 7 & 6 \\ 3 & 7 & 4 & 1 \\ 6 & 0 & 7 & 10 \end{pmatrix}$$

Claire Kim:

Let A be a diagonalizable matrix whose only eigenvalues are 2 and 3. Then what is $5A - A^2$?

David Liang:

Find the algebraic and geometric multiplicity for every eigenvalue of the following matrix:

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & -2 \\ 2 & 0 & 0 \end{pmatrix}$$

Elaine Wang:

Let A be a square matrix and its characteristic polynomial is given by

$$p(t) = (t-1)^2(t-2)^3(t-3).$$

Find the rank of A.

Emilio Peláez

An inner product can be written as $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^T M\mathbf{x}$ for some matrix M. Show that in $\mathbb{R}^3$, if the matrix M is a permutation matrix, the inner product with the form given is well-defined.

Emmanuel Abayev:

Suppose that $\mathbf{v}$ is an eigenvector of A^2. Is it necessarily true that $\mathbf{v}$ is also an eigenvector of A?

Evan Mathura:

Determine, without computation, whether or not the matrix
$$\begin{pmatrix} 4 & 0 & 8 \\ 0 & -2 & -10 \\ 0 & 0 & -12 \end{pmatrix}$$
is diagonalizable. If it is, then diagonalize it.

Everett Smith:

Suppose A is a permutation matrix. What are the possible eigenvalues and eigenvectors for A? When is A diagonalizable?

Felix Farb:

Find an example of two diagonal matrices with the same determinant and the same trace which do not have the same eigenvalues.

Frank Ragone:

What happens to the eigenvalues and eigenvectors of this matrix (in $\mathbb{C}$) if you scale its first row by three? What about scaling its second row by i? $\begin{pmatrix} 3 & 4 \\ 2 & 8 \end{pmatrix}$

Graydon Schulze-Kalt:

Let A be a square matrix with eigenvalues $\lambda_1 = 2$, $\lambda_2 = 2$, and $\lambda_3 = 4$, and corresponding eigenvectors $v_1 = (1, 0, 1)^T$, $v_2 = (0, 1, 1)^T$, and $v_3 = (1, 1, 2)^T$.

1. Find a basis for the eigenspace of A corresponding to the eigenvalue $\lambda = 2$.

2. Find a basis for the eigenspace of A corresponding to the eigenvalue $\lambda = 4$.

3. Show that the matrix A is diagonalizable and find a diagonal matrix D and invertible matrix P such that $A = PDP^-1$.

Han Zhang:

Do two unique 2×2 matrices (A, B) exist such that $\det(A) = \operatorname{tr}(A)$, $\det(B) = \operatorname{tr}(B)$, and $\det(AB) = \operatorname{tr}(AB)$?

Illia Voloshyn:

Does there always exist a matrix A_k such that $\forall k > 1$ we have $\det(A_k^k) = k \det(A_k)$? Explain your answer.

Jacky Chen:

For an invertible linear operator A in a finite-dimensional vector space V, prove that if T is diagonalizable, then T^{-1} is also diagonalizable.

Jonathan Norber:

Calculate the eigenvalues and eigenvectors for the following matrices over $\mathbb{C}$:

$$\begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix} \qquad \begin{pmatrix} i & i \\ i & i \end{pmatrix}$$

Can you come up with an explanation as to why these eigenvectors would be eigenvectors for this matrix?

Juliet Oliver:

Consider the vector space of continuous functions on $[0, 1]$ over $\mathbb{R}$. Give three meaningfully different examples (e.g. not scalar multiples of each other) of functions for which the derivative transformation has at least one eigenvector and corresponding eigenvalues. Then, give three examples of functions that have no eigenvectors. What, if any, properties do the functions have in each group?

Justin Tarquino:

The two shear matrices $\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ y & 1 \end{pmatrix}$ can be multiplied in that respective order such that the determinant of their product is still 1. Calculate the eigenvalues and eigenvectors of $\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ y & 1 \end{pmatrix}$.

Justin Wong:

Show that the operator $T \in L(\mathbb{C}^\infty)$ defined by $T(x_1, x_2, ...x_n) = (0, 0, 0, x_1, x_2, ...)$ has no eigenvalues.

Maanav Kaistha:

Find a general formula for the area of a parallelogram with the points $(x_1, y_1), \ldots, (x_n, y_n)$ (Hint: start with three coordinates and see if you can notice a pattern as you increase your number of points).

Mariam Contractor:

Let A, B be $n \times n$ matrices. Prove that λ is an eigenvalue of AB if and only if it is an eigenvalue of BA. Do their eigenvectors also coincide?

Mark O'Shea:

i) Give a few examples of a 3×3 matrix with an algebraic multiplicity equal to its geometric multiplicity.

ii) Give a few examples of a 3×3 matrix with an algebraic multiplicity greater than its geometric multiplicity. What condition(s) would such a matrix need to satisfy?

Marten Garicano:

Compute the matrix which reflects any vector in $\mathbb{R}^2$ across the line $y = 2x$.

Michael Clancy:

Consider the matrix $A = \begin{pmatrix} 3 & -1 & 2 & 0 \\ -2 & 1 & 3 & 0 \\ 1 & 2 & -1 & -3 \\ -1 & 0 & -2 & 4 \end{pmatrix}$.

If possible, calculate a diagonalization B of A and provide the invertible matrix P for which $B = PAP^{-1}$. If the matrix is not diagonalizable, prove why this is the case.

Mikhail Gebeyehu:

For any space of polynomials, is an integral function of a polynomial in space P_n to P_{n+1} a linear transformation? Show why or why not.

Myra Singh:

Let A be a matrix such that $A^2 = A$. Find all possible eigenvalues of A.

Oliver Bock:

Consider the eigenvalue $\lambda_1 = 4$ corresponding to the eigenvector

$$\begin{pmatrix} \frac{1}{6} \\ 1 \end{pmatrix}$$

and $\lambda_2 = 10$ corresponding to the eigenvector

$$\begin{pmatrix} \frac{2}{3} \\ 1 \end{pmatrix}.$$

Given this information, use diagonalization to calculate the matrix A for these eigenvalues and eigenvectors. Now, consider A^T. Calculate the eigenvalues and eigenvectors of the transpose. More generally, prove that the eigenvalues of the transpose of a matrix will always be the same as those of the original matrix.

Parth Wokhlu:

Diagonalize the following matrix: $\begin{pmatrix} 1 & 0 & 46 \\ 1 & 1 & 1 \\ 23 & 2 & 1 \end{pmatrix}$

Piper Kraske:

Suppose a 2×2 matrix has trace 6 and determinant -4. What are its eigenvalues?

Richard Chen:

Let A be an operator from V to V where V is a 2-dimensional vector space over the complex field. Let A have the eigenvalues 0 and 1 and the corresponding eigenvectors $(4, 9)^T$ and $(16, 25)^T$. Give the matrix of A in the standard basis $[A]_S$. Determine if $[A]_S$ is unique among all matrices of A in any basis of V.

Shivant Krishnan:

Always, Sometimes, Never: Two similar matrices have equal determinants.

Simon Seignourel:

Give an example of a 3×3 matrix with only one nonzero eigenvalue. Is this matrix diagonalizable? Why or why not?

Syris Shao:

Prove that the set of eigenvectors of a matrix A form a vector space.

Yazid Badawy:

Diagonalize: $\begin{pmatrix} 5 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$

Zane Miller:

Prove the following statement:

The closed-form solution of the recurrence

$$F_n = 2F_{n-1} - F_{n-2} + 2F_{n-3}$$

is

$$F_n = F_0 i^n + F_1(-i)^n + F_2 2^n.$$